

Assignment 1: Docker

1. Nginx is one of the most popular web servers. Container images are readily available on Docker Hub. Write down the commands needed to:
 - a. Download the images tagged 1.29.5 and 1.29.5-alpine locally.
 - b. Compare the sizes of the two images.
 - c. Start `nginx:1.29.5` in the background, with the appropriate network settings to forward port 80 locally at port 8000 and use a browser (or `curl` or `wget`) to see that calls are answered. What is the answer?
 - d. Confirm that the container is running in Docker.
 - e. Get the logs of the running container.
 - f. Stop the running container.
 - g. Start the stopped container.
 - h. Stop the container and remove it from Docker.
2. Following the previous exercise, start the Nginx container again and issue the commands needed to:
 - a. Open a shell session inside the running container and change the first sentence of the default page to "Welcome to MY nginx!". Close the session. Validate the change.
 - b. From your computer's terminal (outside the container) download the default page locally and upload another one in its place. Validate the change.
 - c. Close the container, delete it and start another instance. Do you see the changes? Why?Create a simple HTML page in a local folder and issue the commands needed to:
 - d. Start an Nginx container to serve the page from the local folder instead of the default page. Validate that the correct content is served.
3. An example Django application is available in the course's repository on GitHub (<https://github.com/chazapis/hy548>, in the `django` folder). Extend the included `Dockerfile`. Make sure that the database is kept in a separate volume. Also include `vi`, provided by package `vim-tiny`, in case someone needs to inspect or edit a file in a running container. Provide the new `Dockerfile` and the commands needed to:
 - a. Create the container image. Explain how much bigger your own image is compared to the image you were based on. Why?
 - b. Start the extended container twice, one with "debug mode" enabled and one without (set `DJANGO_DEBUG=0`). Validate that both are running and serving different content. What do you see?

- c. Upload the image to your Docker Hub account.
4. Now that you have the Django application with the corresponding Dockerfile in your GitHub repository and a Docker Hub account, create a GitHub Action that will automatically build and push the image (the workflow should be initiated by the user). Provide the YAML of the workflow you made.

Notes:

- The assignment is personal.
- All exercises contribute equally to the overall grade (unless individual percentages are defined).
- Ask for questions and clarifications at the beginning of lectures or via email.
- Write down your answers in a Markdown-formatted text file in either Greek or English.
- Place all files in a folder created for the assignment. Submit your answers in the private GitHub repository that you will use for the course. Share the repository with the instructor (username "chazapis").
- Commits done after the exercise's deadline will be ignored.